

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SUBJECT: Computer Vision

■ CS 6220 ■ Duration 120 minutes ■ Full Marks 30 ■ Date: 28-October-2021■

*After Completing the answer you need to mail me the digital copy of the answer sheet.
Number within bracket indicate full mark for that question.*

1. (a) A greyscale transformation can be applied directly onto a greyscale image to manipulate its pixel values (assuming the range is $[0,255]$). Draw the diagrams for the following greyscale transformations: [3]
 - thresholding the image at pixel value 100.
 - linearly stretch the intensity in the interval $[100,200]$ to $[0,255]$.(b) When would it be suitable to use histogram equalization? Briefly outline the steps to histogram-equalize a given image. [3]
2. (a) Let $f(x)$ be a 1-dimensional continuous function. Let $F(\omega)$ be its Fourier transform. [3]
 - Write down the expression for $F(\omega)$ as a function of f .
 - Write down the inverse Fourier transform of $F^{-1}(\omega)$.(b) With suitable diagram compare the human visual system and computer vision system. [3]
3. (a) Discuss various edge detection methods. [3]
(b) Specify the 3×3 Prewitt operators used for edge detection. Sketch the result when one of these masks is applied to the greyscale image below. You only need to compute the values for the central region of the image that is fully covered by the mask. [3]

7	8	8	9	9	10
7	8	9	35	39	40
8	9	10	37	39	40
8	7	37	36	38	41
7	7	9	37	19	40
4	8	9	36	21	40

4. (a) What is image segmentation? Discuss the two different approaches for region oriented segmentation with suitable example. [3]
(b) What are Unimodal, Bimodal and Multi-modal histograms? Explain the optimal **Otsu thresholding** algorithm. [3]
5. Cluster the following eight points (with (x,y) representing locations of the data point) using k -means clustering algorithm. $A_1(2,9)$, $A_2(4,5)$, $A_3(6,9)$, $A_4(3,9)$, $A_5(7,5)$, $A_6(1,2)$, $A_7(8,4)$, $A_8(2,5)$. Consider the initial points considering $k = 3$ are $A_1(2,9)$, $A_3(6,9)$, and $A_6(1,2)$. The distance function between two points $a(x_1, y_1)$ and $b(x_2, y_2)$ is defined by $d(a, b)$, [6]

$$d(a, b) = |x_2 - x_1| + |y_2 - y_1| \quad \text{for } x_1 \neq x_2 \text{ and } y_1 \neq y_2$$
$$= 0 \quad \text{otherwise}$$

*****Wish you all the best*****